

1. What is phishing? Give an example.

Definition

Phishing is a type of cyber attack in which an attacker pretends to be a trusted person or organization (such as a bank, company, or government agency) to deceive users into sharing sensitive information like usernames, passwords, credit card numbers, or OTPs.

It is a form of **social engineering attack**, where the attacker exploits human trust rather than technical weaknesses.

Key Features of Phishing

- Uses **fake emails, messages, or websites**
- Creates a sense of **urgency or fear** (e.g., account blocked)
- Appears to come from a **trusted source**
- Aims to **steal confidential data**

Types of Phishing (Brief)

- **Email Phishing:** Fake emails sent to many users
- **Spear Phishing:** Targeted attack on a specific person
- **Smishing:** Phishing via SMS
- **Vishing:** Phishing via phone calls

Diagram (Phishing Process)



Working of Phishing (Step-by-Step)

1. The attacker creates a **fake website** similar to a real one (e.g., bank website).
2. A **fraudulent email or message** is sent to the victim.
3. The message contains a **malicious link**.
4. The user clicks the link and is redirected to the **fake website**.
5. The user unknowingly enters **sensitive information**.
6. The attacker collects and misuses the data.

Example

A user receives an email that appears to be from their bank stating, “*Your account will be blocked. Click here to update your details.*”

The link opens a fake website that looks exactly like the bank’s official site. When the user enters their username, password, and OTP, the attacker captures this information and uses it to access the user’s real bank account.

Effects of Phishing

- Financial loss
- Identity theft
- Unauthorized account access
- Loss of privacy

Conclusion

Phishing is one of the most common and dangerous cyber attacks because it targets human behavior. Awareness and careful online practices are essential to prevent such attacks.

2. Explain Types of Computer System Attacks

1. Malware Attacks: Malicious software designed to harm systems.

- Virus
- Worm
- Trojan Horse
- Ransomware

Example:

A file named “*Free Game.exe*” installs a virus that deletes files.

2. Phishing Attacks: Fake emails/websites trick users into giving personal information.

Example:

An email pretending to be from a bank asks for your password.

3. Denial of Service (DoS): Flooding a server with traffic so it crashes.

Example:

A website receives millions of fake requests and stops working.

4. Password Attacks: Trying to guess or steal passwords.

Example:

Using software to try thousands of password combinations.

5. Man-in-the-Middle Attack: Attacker secretly intercepts communication.

Example:

Hacker intercepts data on public Wi-Fi.

6. Insider Attacks: Employees misuse access.

Example:

An employee steals company data.

3. Explain the Motives of Computer Crimes.

◆ Introduction

Computer crimes (cyber crimes) are illegal activities carried out using computers or the internet. These crimes are committed for various reasons depending on the attacker's goals and intentions.

◆ Motives of Computer Crimes

◆ 1. Financial Gain

The most common motive is to earn money illegally.

- Stealing bank details, credit card information
- Online frauds and scams

👉 Example:

A hacker steals credit card details and uses them for online shopping.

◆ 2. Revenge

Some individuals commit cyber crimes to take revenge on organizations or people.

- Disgruntled employees
- Personal conflicts

👉 Example:

An ex-employee deletes important company data after being fired.

◆ 3. Curiosity or Challenge

Some attackers, especially beginners, hack systems just to test their skills or for fun.

- No intention of financial gain

- Desire to prove technical ability

 Example:

A student hacks a website just to see if they can break its security.

♦ **4. Political or Social Motives (Hacktivism)**

Cyber attacks are carried out to promote political or social causes.

- Defacing websites
- Spreading messages

 Example:

A group hacks a government website to protest against policies.

♦ **5. Espionage (Spying)**

Stealing confidential or sensitive information for competitive or national advantage.

- Industrial espionage
- Government spying

 Example:

Hackers steal secret business plans of a company.

♦ **6. Terrorism**

Cyber attacks aimed at creating fear or disrupting critical systems.

- Attacks on infrastructure
- National security threats

 Example:

Cyberattack on power grids or defense systems.

♦ **7. Recognition or Fame**

Some hackers want to gain popularity or recognition.

- Showing off hacking skills
- Gaining attention in hacker communities

 Example:

A hacker defaces a popular website and leaves their name.

Conclusion

Computer crimes are driven by various motives such as financial gain, revenge, curiosity, and political intentions. Understanding these motives helps in developing better security measures and preventing cyber attacks.

4. Explain Denial of Service.

◆ Definition

A Denial of Service (DoS) attack is a type of cyber attack in which an attacker attempts to make a computer system, server, or network unavailable to legitimate users by overwhelming it with a large number of requests or traffic.

◆ Explanation

In a DoS attack, the attacker sends excessive fake requests to a server. Since the server has limited resources (like memory and bandwidth), it becomes overloaded and cannot respond to genuine users. As a result, the system slows down or completely crashes.

◆ Types of DoS Attacks

1. DoS (Denial of Service):

- Attack from a single system

2. DDoS (Distributed Denial of Service):

- Attack from multiple systems (botnets) simultaneously
- More powerful and difficult to stop

Diagram (Working of DoS Attack)



◆ **Example**

An e-commerce website receives millions of fake requests during a sale. The server becomes overloaded and crashes, preventing real customers from accessing the website.

◆ **Effects of DoS Attack**

- Website or service downtime
- Loss of revenue and business
- Customer dissatisfaction
- Damage to reputation

◆ **Prevention Measures**

- Use firewalls and intrusion detection systems
- Limit traffic using rate limiting
- Use load balancers
- Monitor network traffic regularly

Conclusion

A Denial of Service attack disrupts the availability of online services by overloading systems. Proper security measures are essential to detect and prevent such attacks.